Team 3

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Improving Eyewear Demand forecasting for VSP Vision

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Content overview

Who is VSP Vision?

- Founded in 1955 by optometrists, it functions as a vision benefits company rather than a traditional insurance company
- Network base of over 85 million members and 39k doctors
- 5 eyewear solutions for different customer choices and preferences
- Eyeconic is mostly focused on providing “fashionable” options for the daily user

eyeconic

vsp optics

MARCHON
a vsp vision company

ALTAIR
a vsp vision company

ALLURE
a vsp vision company

Brief Overview of the Problem

The Business Challenge

New Frames do not have historic data to rely on for inventory data, hence we have to get creative to forecast demand

Why it matters?

Over Forecasting → excess inventory & markdown risk
Under Forecasting → stock outs & lost sales

Our Objective

Develop a demand forecasting model to estimate initial order quantities for these new frames to avoid these problems

Market Analysis

Understanding the eyeglasses market

Market Structure and Competitive Positioning

- U.S. vision insurance market is highly concentrated, dominated by a few national players
- Employer-sponsored plans drive the majority of enrollment
- Competition occurs across networks, pricing, and eyewear partnerships
- Market led by a small group of insurers with large provider networks
- Competitors often integrate benefits with broader healthcare or eyewear ecosystems
- Increasing overlap between insurance providers and vertically integrated optical companies

Competitors in the eyeglasses market

- Essilor Luxottica
- Warby Parker
- Independen t retailers

Competitors in the Vision Insurance market

- Davis Vision
- EyeMed
- United Healthcare

Stakeholders

	VSP Vision	Consumers	Optometrists /Providers	Suppliers	Retailers and Distributors
PRIMARY GOAL	Optimal inventory levels	Style, affordability, convenience	Patient satisfaction & practice efficiency	Stable production & predictable orders	Sales performance & turnover
INFLUENCE ON DEMAND	Inventory availability shapes which frames customers ultimately purchase	Preferences determine brand, style, and purchase frequency	Provider recommendations can influence where consumers buy frames from.	Supplier lead times restrict replenishment and assortment flexibility.	Store assortment differences create regional demand variation.

Consumer Analysis

Standard Consumer Journey

Need Recognition

Changes in vision, annual benefits renewal, or lifestyle/fashion needs trigger purchase intent

Eye Exam and Prescription

Eye exams usually act as the primary purchase trigger, determining when customers enter the buying cycle.

Frame Consideration

Consumer evaluates style, brand, comfort, price and recommendations from providers on available assortment options

Purchase Decision

Selection is influenced by style trends, insurance benefits, availability, and in-store experience

Usage and Replacement cycle

Future purchases follow predictable replacement cycles tied to prescription updates and frame lifecycle.

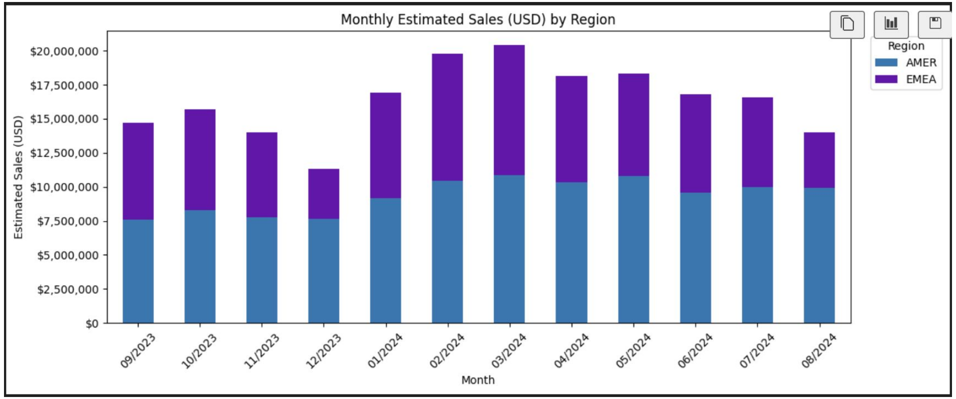
High-level Trends

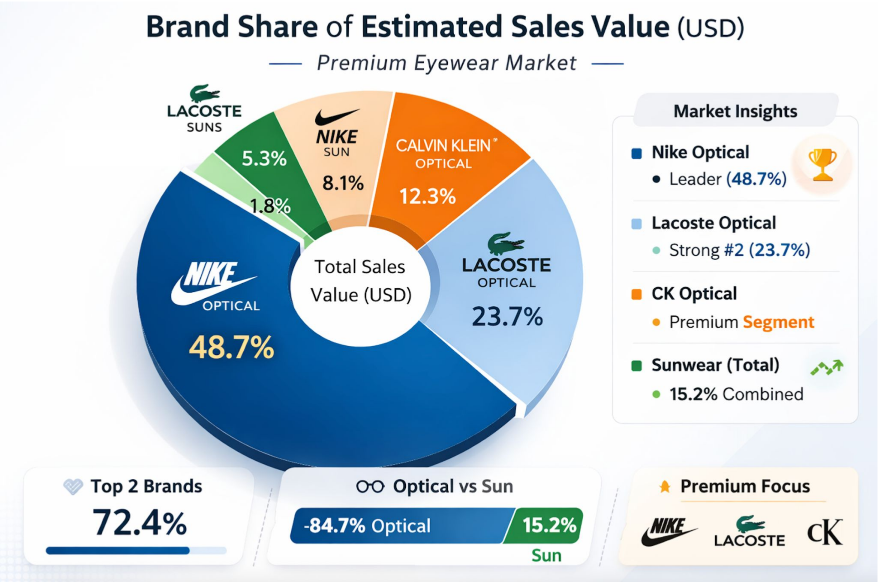
Most popular frame in EMEA

Lacoste 2707

Most popular frame in AMER

NIKE 7125





Key Optical Consumer Segments

Luxury Lifestyle Statement

Targets image-conscious, low price-sensitivity consumers who view eyewear as a premium style and status accessory.

Fashion-Forward Acetate

Appeals to style-driven optical buyers who prioritize modern design, silhouette, and everyday visual impact

Comfort-Driven Premium

Designed for consumers seeking premium eyewear centered on long-wear comfort, flexibility, and durability rather than bold fashion.

Core Lifestyle Essentials

Targets practical, value-conscious consumers seeking reliable, lightweight frames for daily use across multiple settings

Key Sunglasses Consumer Segments

Accessible Sport Lifestyle Sun

Targets casual, active consumers looking for comfortable, durable sunglasses

Fashion Acetate Sunwear

Appeals to fashion-oriented buyers who treat sunglasses as accessories and prioritize aesthetics over performance features.

Value Performance Sunwear

Targets price-sensitive, active consumers seeking functional, sport-inspired sunglasses.

Elite Performance Shield Sun

Designed for performance-driven consumers who prioritize technical features, coverage, and lightweight durability..

What are the key features that drive frame sales?

- Segmentation reduces forecasting risk

  Grouping frames into clusters with similar features allows demand to be estimated at the segment level rather than per individual frame.

- Material and color are the strongest sales driver

  Clusters are primarily differentiated by acetate, bio-injected plastics, and titanium, indicating that material strongly shapes consumer preference, perceived value, and price tolerance.

- Comfort features enable premium positioning

  Bio-injected frames combined with silicone/ergonomic elements successfully appear in premium optical clusters, suggesting comfort is a key upsell factor.

- Sunwear demand splits into fashion vs performance

  Clusters reveal two distinct drivers: style-led acetate sunwear and function-led high base curve/performance frames, requiring differentiated assortment strategies.

Our Methodology

From Raw Data to Demand Predictions

We built a machine learning model spanning 12 months of sales data across 3 brands and 162 SKUs — transforming transactional records into forward-looking demand signals.

12	162	3
Months of Data	**Total SKUs**	**Brands**
Sep 2023 through Aug 2024	Across optical and sunglass categories	Nike · Lacoste · Calvin Klein

Key Data Assumptions

1. **Product Attributes Inferred**
   Style codes in ATPs lacked full attribute data. Frame shape, material, and gender were inferred using Eyeconic, Amazon, and industry naming conventions.

2. **Materials Assigned by Pattern**
   Where material was missing, it was assigned based on known brand conventions — e.g. Nike Bio Inj-G820 for sport optical lines, Acetate for Lacoste lifestyle codes.

3. **Premium Collections Tagged**
   Titanium and premium metal frames were identified and tagged separately rather than grouped into generic "Metal," preserving price and performance signal fidelity.

4. **Euro Pricing Excluded**
   EMEA retail prices in EUR were unavailable. All price analysis uses US retail pricing only. EMEA price signals may differ from those presented.

5. **Wholesale Estimated at Retail ÷ 2**
   Where wholesale price was missing, it was estimated using the standard optical industry convention: wholesale ≈ 50% of retail. Used only in margin ratio features.

6. **Missing Prices Approximated**
   A small number of SKUs lacked retail price. These were assigned the median price for their brand and category tier.

Full methodology, including attribute inference logic and data cleaning rules, is documented in the accompanying analysis report.

How We Prepared the Data for Modeling

1

2

01 · Cleaning & Standardizing

Fixed size typos (3.0→53mm), standardized text casing across all categorical columns, filled missing color/material fields from a second attributes table.

02 · Reshaping to Long Format

Pivoted 12 monthly sales columns into one row per SKU-Region-Month, giving the model a time-series structure — each row equals one observation.

3

4

03 · Feature Engineering

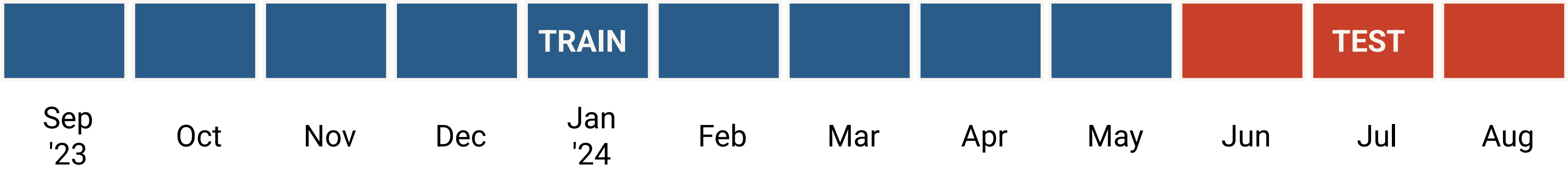
Built 13 features: lag features (1, 2, 3, 6 months), rolling averages, expanding max, cumulative sales, MoM growth rate, brand context, and SKU-vs-brand ratio.

04 · Log-Transform Target

Sales was right-skewed — a few SKUs sold 1,000+/month while most sold ~50. Log(Sales+1) compressed outliers so the model could learn across the full range.

Key decision: Dropped ColorDescription (too many unique values) and StyleCode (would cause data leakage — the model would memorize SKU IDs instead of learning patterns).

Rigorous Train / Test Split



No data leakage. Future months never seen during training — model has to genuinely predict the last 3 months.

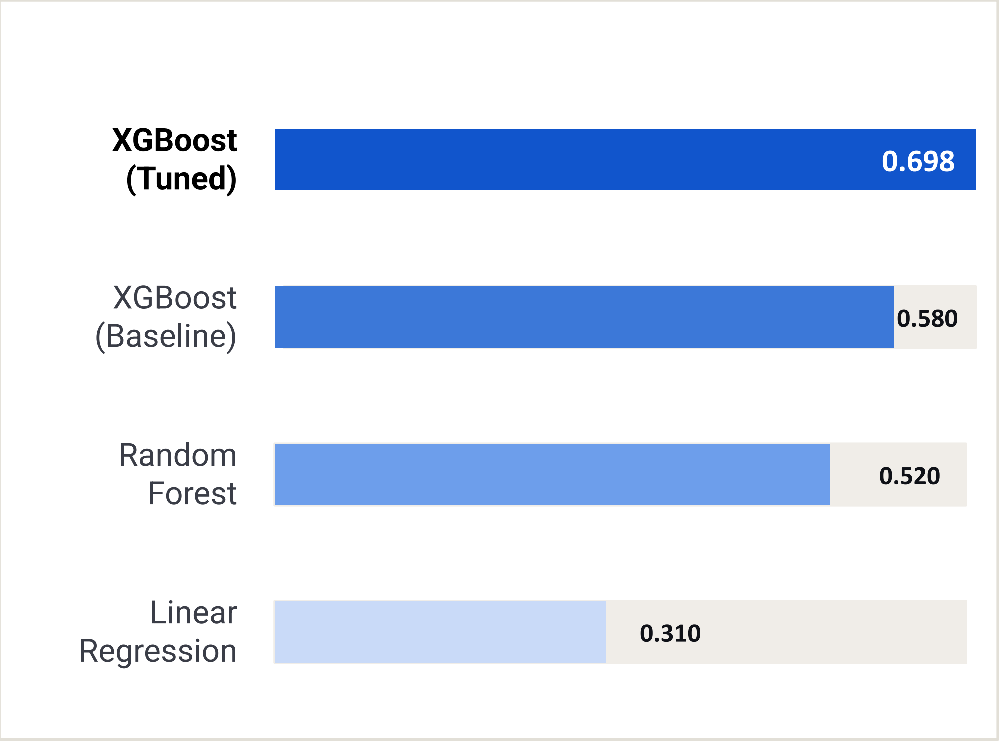
3-fold cross-validation. Tuning scores validated across multiple data cuts, not just a single lucky split.

Dual R^2 reporting. Log-scale (0.648) for model fit; raw units (0.442) for business accuracy. We report both honestly.

MAE of 29 units/month. For SKUs averaging 50–300 units/month, this is actionable for purchasing decisions.

We Compared 3 Models — XGBoost Won

R^2 Score (Log Scale) — Higher is Better



Why XGBoost?

→ Handles non-linear relationships — sales doesn't move in straight lines

→ Native support for mixed data types (numbers + categories after encoding)

→ Built-in regularization prevents overfitting on our 162-SKU dataset

→ Gradient boosting: each tree corrects the errors of the last

The Model Results: What Drives Frame Sales

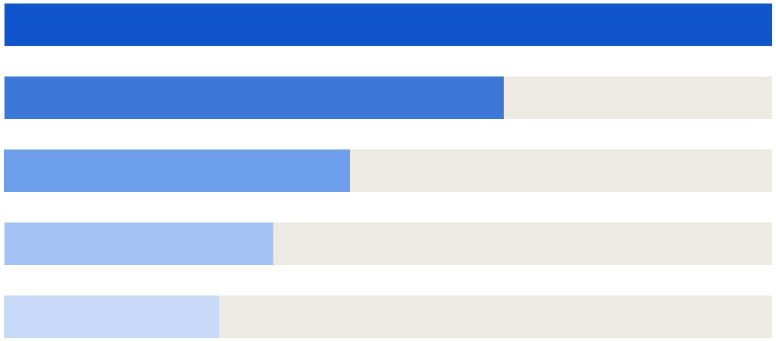
0.695
R² (Log Scale)
Primary metric · explains 65% of demand variance

0.542
R² (Raw Units)
Harder benchmark due to high-volume outliers

26.4
MAE (Units/month)
Avg prediction off by ~29 units per SKU/month

33
Input Features
Lag, rolling avg, brand, region, category & more

What the Model Says Matters Most:



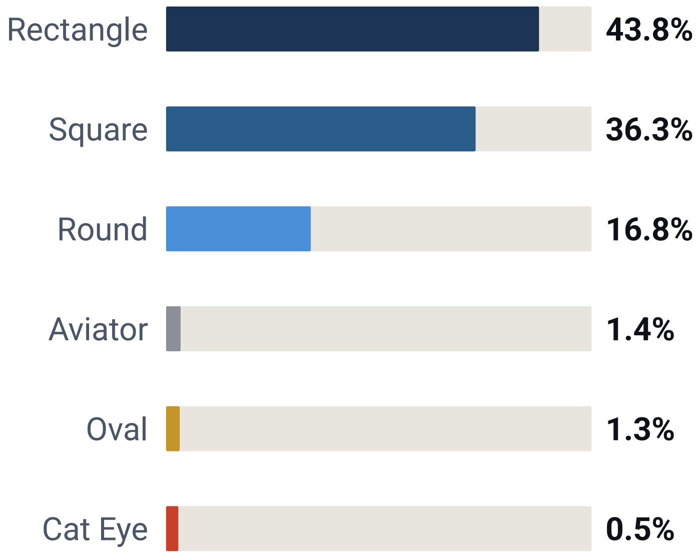
Rolling 6-Month Avg Sales — #1
SKU Historical Max Sales — #2
SKU Expanding Avg Sales — #3
Brand (Nike / Lacoste / CK) — #4
Optical vs Sunglass — #5

Key insight:
Past sales patterns explain demand better than any product attribute. Brand matters more than shape, material, or size.

WHAT PATIENTS BUY

Classic shapes dominate — but a few niches are undersupplied

SHARE OF PREDICTED DEMAND BY SHAPE



90% of demand is Optical frames. Sunglasses are 10% and declining.

82% is Rectangle or Square

Visionworks patients want classic, professional frames. This is stable and should anchor the assortment.

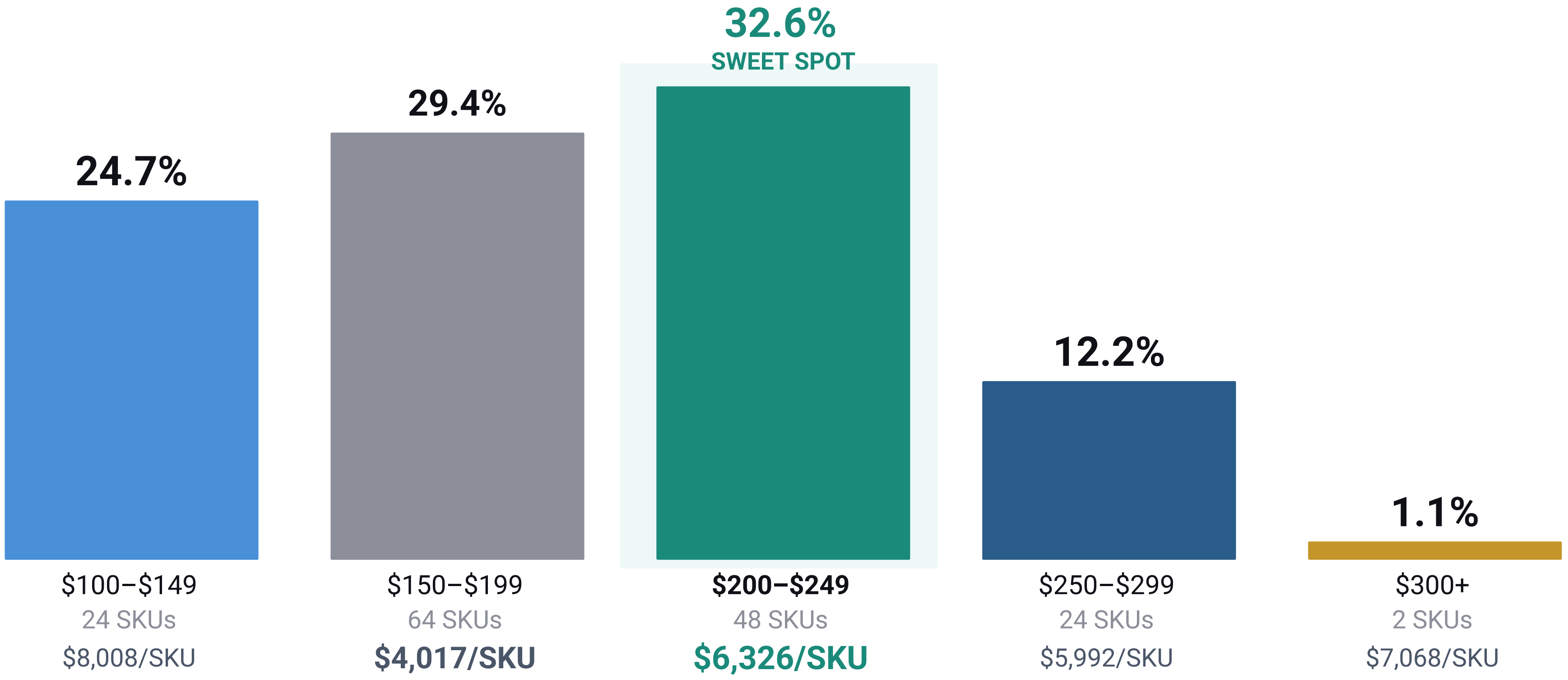
Oval: 1 SKU, fully proven

One Oval style generated 9,400 units. This is the most obvious white space in the entire assortment.

Cat Eye: trending, not stocked

Only 2 Cat Eye SKUs exist. The style is growing nationally — Visionworks is behind.

The $200–$249 band is where patients spend — and where efficiency is highest



REGIONAL ANALYSIS

AMER and EMEA are not the same market — don't buy them the same way

AMER · 56% of demand · 133,765 units	
#1 Brand	**Nike (61% of AMER)**
#2 Brand	**Calvin Klein (19%)**
#3 Brand	Lacoste (16%)
Top Size	**54mm** — broader faces
Price Mix	Heavy in $150–$199 tier
Sunglass	Low — prescription-first market

EMEA · 44% of demand · 104,434 units	
#1 Brand	**Lacoste (39% of EMEA)**
#2 Brand	**Nike (33%)**
#3 Brand	Lacoste Sun (14%)
Top Size	**53mm** — smaller, fitted
Price Mix	Stronger $200–$249 pull
Sunglass	Higher — fashion-driven

DEMAND FORECAST

238,199 units predicted — here is where they come from

BRAND	SHARE OF FORECAST	UNITS (4MO)	AVG/MO/SKU
Nike Optical	51.3%	122,266	72.6
Lacoste Optical	23.3%	55,411	54.8
CK Optical	15.2%	36,156	40.9
Lacoste Sun	7.4%	17,609	37.0
CK Sun	2.1%	5,012	11.8
Nike Sun	0.7%	1,745	36.4
TOTAL FORECAST — ALL BRANDS		**238,199 units**	

BUY LIST

The top 10 SKUs to prioritise — by predicted 4-month volume

#	SKU	Brand	Region	Size	Colour	Pred/mo	4mo Qty	Priority
1	**NIKE 7280**	Nike	AMER	50mm	Gunsmoke	324	**1,294**	**HERO**
2	**NIKE 7280**	Nike	AMER	50mm	Clear	314	**1,255**	**HERO**
3	**L2707**	Lacoste	EMEA	53mm	Blue Steel	297	**1,186**	**HERO**
4	**NIKE 7240**	Nike	AMER	55mm	Grey	262	**1,047**	**HERO**
5	**NIKE 7287**	Nike	AMER	54mm	Matte D	251	**1,003**	**HERO**
6	**NIKE 8054**	Nike	AMER	55mm	Satin G	229	**915**	**HIGH**
7	**NIKE 7160**	Nike	AMER	55mm	Black/C	218	**871**	**HIGH**
8	**NIKE 7160**	Nike	AMER	55mm	Crystal	215	**860**	**HIGH**
9	**NIKE 7240**	Nike	AMER	53mm	Grey	201	**803**	**HIGH**
10	**NIKE 4315**	Nike	AMER	55mm	Satin N	198	**794**	**HIGH**

Recommendations

Customer persona analysis

- Add segment signals: age band, plan type, channel (online vs store), repeat vs first-time
- Model segment-level preferences: color families, size curves, sun vs optical mix
- Use segment forecasts to tailor initial order depth by “who buys what”

Expand geographic granularity (country/market level)

- Replace AMER/EMEA with country/market clusters + local demand baselines
- Allocate inventory by market-level “hero” colors/sizes to reduce misallocation
- Capture regional seasonality differences (weather, fashion, benefit timing)

Expand product attribute richness (metadata)

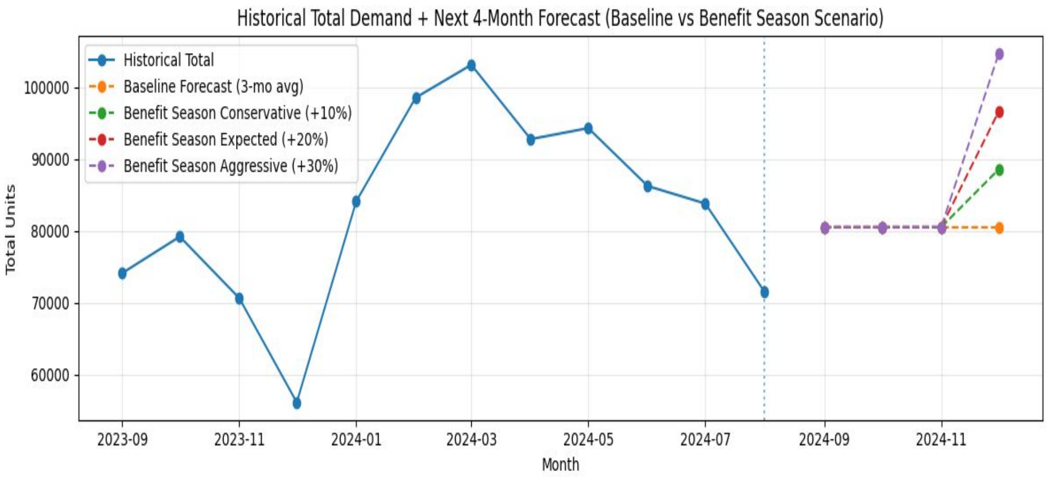
- Add launch/marketing flags: campaign exposure, featured placement, email/push timing
- Add commercial context: price tier changes, discounts, channel placement
- Add post-purchase signals: returns/fit issues to explain drops and improve forecasts

Integrate benefit renewal timing

- Add features: benefit reset month, open enrollment window, "days-to-expiry"
- Create surge flags for the last 30–60 days before reset (demand pull-forward)
- Pre-position inventory and coordinate promos only for in-stock winners

VSP Vision

"Use-It-or-Lose-It" Season
A benefit deadline demand play



Treat benefit reset and enrollment windows as predictable demand surges, planning inventory and messaging in coordination.

Why Now

Benefit-driven purchasing is time-bound unplanned spikes create stockouts and rushed fulfillment, leading to lost revenue.

How to Implement

- Build a Benefit Deadline calendar (December + open enrollment windows)
- Pre-position inventory for forecasted winners: hero colors, top sizes, optical vs sun split
- Coordinate marketing offers with inventory readiness (avoid promoting items at risk of stockout)

Use VSP Vision Care to run a benefits, expiration program

Thank you!

Questions?